20260929 Quiz

for (MATH301) Modern Algebra I

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Exercise 1.7.1. Let F be a field. Show that $F^\times = F \setminus \{0\}$ acts on F by $g \cdot a = ga$. State which field axioms are used.

Solution. For $g, h \in F^\times$ and $a \in F$, associativity of multiplication gives

$$g \cdot (h \cdot a) = g(ha) = (gh)a = (gh) \cdot a.$$

The multiplicative identity satisfies $1 \cdot a = a$. Thus the two action axioms hold. They use precisely the associativity and identity axioms for multiplication. The other field axioms ensure that nonzero elements form the group $F^\times$: in particular, multiplication is closed on nonzero elements and every nonzero element has a multiplicative inverse.

□

Exercise 1.7.16. Let G act on itself by conjugation, $g \cdot a = gag^{-1}$. Show that this defines a left group action.

Solution. For $g, h, a \in G$,

$$g \cdot (h \cdot a) = g(hah^{-1})g^{-1} = (gh)a(gh)^{-1} = (gh) \cdot a.$$

Moreover $1_G \cdot a = 1_G a 1_G^{-1} = a$. Both defining axioms are satisfied, so conjugation is a left action of G on itself.

□

Exercise 1.7.17. For $g \in G$, prove that $c_g : G \to G$, $x \mapsto gxg^{-1}$, is an automorphism. Deduce that $|x| = |gxg^{-1}|$ and $|A| = |gAg^{-1}|$ for any $A \subseteq G$.

Solution. For $x, y \in G$,

$$c_g(xy) = gxyg^{-1} = (gxg^{-1})(gyg^{-1}) = c_g(x)c_g(y).$$

Thus c_g is a homomorphism. Since $c_{g^{-1}} \circ c_g = c_g \circ c_{g^{-1}} = \mathrm{id}_G$, it is bijective, hence an automorphism. An isomorphism preserves element orders because $c_g(x)^m = 1$ if and only if $x^m = 1$. Its restriction $A \to gAg^{-1}$ is also a bijection, proving $|A| = |gAg^{-1}|$.

□

Exercise 2.1.2. For each subset below, prove that it is not a subgroup of the indicated group.

(a) The transpositions in S_n, $n \geq 3$.

(b) The reflections in D_n, $n \geq 3$.

(c) If $n > 1$ is composite and G has an element of order n, the set $\{x \in G : |x| = n\} \cup \{1_G\}$.

(d) The odd integers together with 0 in $(\mathbb{Z}, +)$.

(e) The set $\{x \in \mathbb{R} : x^2 \in \mathbb{Q}\}$ in $(\mathbb{R}, +)$.

Notation. As in MATH000, D_n is the symmetry group of a regular n-gon, of order $2n$ (written D_{2n} in the alternative convention of Dummit–Foote).

Solution.

(a) The identity permutation is not a transposition; hence the set does not contain the identity.

(b) The identity is a rotation, not a reflection, so the set does not contain the identity.

(c) Take $x \in G$ of order n, and write $n = ab$ with $1 < a, b < n$. Then x belongs to the set, but $|x^a| = b$, so x^a does not. Any subgroup containing x would contain all its powers.

(d) We have 1 in the set but $1 + 1 = 2$ is not, so it is not closed under addition.

(e) Both 1 and $\sqrt{2}$ belong to the set, since their squares are rational, but $(1 + \sqrt{2})^2 = 3 + 2\sqrt{2}$ is irrational. The set is not closed under addition.

□

Exercise 2.1.3. Show that $H = \{1, r^2, s, sr^2\}$ and $K = \{1, r^2, sr, sr^3\}$ are subgroups of $D_4 = \langle r, s \mid r^4 = s^2 = 1,\ srs = r^{-1} \rangle$.

Solution. Both r^2 and s are involutions and commute, since $sr^2s^{-1} = r^{-2} = r^2$. Therefore

$$H = \langle r^2, s \rangle = \{1, r^2, s, sr^2\},$$

which is a subgroup. Similarly, sr has order 2 and commutes with r^2. Thus

$$K = \langle r^2, sr \rangle = \{1, r^2, sr, r^2sr\} = \{1, r^2, sr, sr^3\},$$

which is also a subgroup. In each case the displayed elements exhaust the subgroup generated by two commuting involutions.

□

Exercise 2.1.6. Let G be abelian. Prove that its elements of finite order form a subgroup (its torsion subgroup). Give a nonabelian example where finite-order elements are not a subgroup.

Solution. Let $T = \{g \in G : |g| < \infty\}$. Certainly $1_G \in T$. If $x, y \in T$ have orders m, n, then commutativity gives

$$(xy^{-1})^{mn} = x^{mn}y^{-mn} = 1_G.$$

Hence $xy^{-1} \in T$, so $T \leq G$.

For a counterexample take

$$\mathbf{A} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \qquad \mathbf{B} = \begin{pmatrix} 0 & 2 \\ 1/2 & 0 \end{pmatrix} \in \mathrm{GL}_2(\mathbb{R}).$$

Both matrices have order 2, whereas $\mathbf{AB} = \mathrm{diag}(1/2, 2)$ has infinite order: its kth power is $\mathrm{diag}(2^{-k}, 2^k) \neq \mathbf{I}$ for every $k > 0$. Thus finite-order elements are not necessarily closed under multiplication in a nonabelian group.

□

Exercise 2.1.13. Let $H \leq (\mathbb{Q}, +)$ and suppose $1/x \in H$ for every nonzero $x \in H$. Prove that $H = \{0\}$ or $H = \mathbb{Q}$.

Solution. Suppose $H \neq \{0\}$ and choose $x = a/b \in H$ with nonzero $a \in \mathbb{Z}$, $b > 0$, and $\gcd(a, b) = 1$. Then $1/x = b/a \in H$. Since H is an additive subgroup, it contains the integers $bx = a$ and $a(1/x) = b$. By Bézout's identity there exist integers u, v such that $ua + vb = 1$, so $1 \in H$. Hence $\mathbb{Z} \subseteq H$.

For any positive integer q, the assumption applied to $q \in H$ yields $1/q \in H$. Multiplication by integers then gives $p/q \in H$ for all $p \in \mathbb{Z}$, so $H = \mathbb{Q}$.

□

Exercise 2.3.1. Find all subgroups of $C_{45} = \langle x \rangle$, give a generator for each, and describe their containments.

Notation. C_n denotes the abstract cyclic group of order n (called Z_n in Dummit–Foote). The additive residue-class group is written $\mathbb{Z}/n\mathbb{Z}$ under MATH000.

Solution. A cyclic group of order 45 has exactly one subgroup of each order dividing 45, namely $H_d = \langle x^{45/d} \rangle$. The six divisors and subgroups are

d	H_d
1	$\{1_G\} = \langle x^{45} \rangle$
3	$\langle x^{15} \rangle$
5	$\langle x^{9} \rangle$
9	$\langle x^{5} \rangle$
15	$\langle x^{3} \rangle$
45	$\langle x \rangle$

For any divisors d, e of 45, uniqueness of cyclic subgroups shows $H_d \leq H_e$ if and only if $d \mid e$. This rule gives all containments; for instance, $H_3 \leq H_9, H_{15}$, while H_5 and H_9 are incomparable.

□

Exercise 2.3.8. Let $C_{48} = \langle x \rangle$. For which integers a does $[1]_{48} \mapsto x^a$ extend to an isomorphism $\mathbb{Z}/48\mathbb{Z} \to C_{48}$?

Solution. The proposed map extends uniquely to $\varphi_a([k]_{48}) = x^{ak}$. It is well defined: changing k by a multiple of 48 changes the exponent by a multiple of 48, and $x^{48} = 1$. It is a homomorphism by the exponent laws. It is bijective exactly when its image contains the generator x, equivalently when x^a generates C_{48}. Since

$$|x^a| = \frac{48}{\gcd(a, 48)},$$

this occurs if and only if $\gcd(a, 48) = 1$. Modulo 48 the solutions are

$$\boxed{1, 5, 7, 11, 13, 17, 19, 23, 25, 29, 31, 35, 37, 41, 43, 47}.$$

□

Exercise 2.3.11. Find all cyclic subgroups of D_4 and exhibit a proper noncyclic subgroup.

Solution. Write $D_4 = \{1, r, r^2, r^3, s, sr, sr^2, sr^3\}$. The rotations have orders $1, 4, 2, 4$, respectively; every reflection sr^j has order 2. Thus the distinct cyclic subgroups are

$$\{1\}, \quad \langle r \rangle, \quad \langle r^2 \rangle, \quad \langle s \rangle, \quad \langle sr \rangle, \quad \langle sr^2 \rangle, \quad \langle sr^3 \rangle.$$

The proper subgroup $H = \langle r^2, s\rangle = \{1, r^2, s, sr^2\}$ is not cyclic: all three of its nonidentity elements have order 2, so none generates its four elements.

□

Exercise 2.3.12. Prove that $(\mathbb{Z}/2\mathbb{Z}) \times (\mathbb{Z}/2\mathbb{Z})$, $(\mathbb{Z}/2\mathbb{Z}) \times \mathbb{Z}$, and $\mathbb{Z} \times \mathbb{Z}$ are not cyclic.

Solution.

(a) Every nonidentity element of $(\mathbb{Z}/2\mathbb{Z})^2$ has order 2, whereas a cyclic group of order 4 would have an element of order 4.

(b) If $(\mathbb{Z}/2\mathbb{Z}) \times \mathbb{Z} = \langle (a, m)\rangle$, then generating $(0, 1)$ forces $km = 1$ for some $k \in \mathbb{Z}$, hence $m = \pm 1$. But every nonzero multiple of (a, m) has nonzero second coordinate, so it cannot generate the nonzero element $(1, 0)$.

(c) If $\mathbb{Z}^2 = \langle (a, b)\rangle$, then generating $(1, 0)$ forces $ka = 1$ and $kb = 0$ for some integer k. Thus $k = \pm 1$ and $b = 0$. All multiples of (a, b) have second coordinate zero, so $(0, 1)$ cannot be generated.

□

Exercise 2.3.19. Let H be a group and $h \in H$. Prove that there is a unique homomorphism $\varphi : \mathbb{Z} \to H$ with $\varphi(1) = h$.

Solution. Define $\varphi(n) = h^n$ for $n \in \mathbb{Z}$. Then $\varphi(m + n) = h^{m+n} = h^m h^n = \varphi(m)\varphi(n)$ and $\varphi(1) = h$. Any homomorphism $\psi : \mathbb{Z} \to H$ with $\psi(1) = h$ must satisfy $\psi(n) = \psi(1)^n = h^n$ for positive, negative, and zero integers n. Therefore $\psi = \varphi$, proving uniqueness.

□

Exercise 2.3.21. Let p be an odd prime and $n \geq 1$. Use the binomial theorem to show

$$(1 + p)^{p^{n-1}} \equiv 1 \pmod{p^n}$$

and, for $n \geq 2$,

$$(1 + p)^{p^{n-2}} \not\equiv 1 \pmod{p^n}.$$

Deduce that $1 + p$ has order p^{n-1} in $(\mathbb{Z}/p^n\mathbb{Z})^\times$.

Hint. Prove the stronger congruence modulo p^{m+2} by induction on m.

Solution. We first show by induction on $m \geq 0$ that

$$(1 + p)^{p^m} \equiv 1 + p^{m+1} \pmod{p^{m+2}}.$$

This is immediate for $m = 0$. Suppose $(1+p)^{p^m} = 1+u$, where $u = p^{m+1} + cp^{m+2}$. The binomial theorem gives

$$(1 + u)^p = 1 + pu + \sum_{j=2}^{p-1} \binom{p}{j} u^j + u^p.$$

Here $pu \equiv p^{m+2} \pmod{p^{m+3}}$. For $2 \leq j \leq p - 1$, the coefficient $\binom{p}{j}$ is divisible by p, so $\binom{p}{j}u^j$ is divisible by $p^{1+j(m+1)}$, hence by p^{m+3}. Also u^p is divisible by $p^{p(m+1)}$, which is a multiple of p^{m+3} because $p \geq 3$. The induction follows.

Taking $m = n - 1$ yields $(1 + p)^{p^{n-1}} \equiv 1 \pmod{p^n}$. When $n \geq 2$, taking $m = n - 2$ yields

$$(1 + p)^{p^{n-2}} \equiv 1 + p^{n-1} \not\equiv 1 \pmod{p^n}.$$

Therefore the order divides p^{n-1} but not p^{n-2}, and so equals p^{n-1}. If $n = 1$, the element is 1 modulo p and its order is $1 = p^0$.

□

Exercise 2.3.24. Let G be finite and $x \in G$.

(a) If $g \in N_G(\langle x \rangle)$, prove that $gxg^{-1} = x^a$ for some $a \in \mathbb{Z}$.

(b) Prove the converse.

Hint. The conjugation map is a bijection; in a finite group, a subgroup contained in another subgroup of the same order must equal it.

Solution.

(a) Since $g\langle x \rangle g^{-1} = \langle x \rangle$, the conjugate gxg^{-1} belongs to the cyclic group $\langle x \rangle$; hence it equals x^a for some integer a.

(b) Suppose $gxg^{-1} = x^a$. Then

$$gx^k g^{-1} = (gxg^{-1})^k = x^{ak} \in \langle x \rangle$$

for every integer k, showing $g\langle x \rangle g^{-1} \subseteq \langle x \rangle$. Conjugation by g is a bijection, so the two finite sets have the same cardinality $|x|$. Hence containment is equality and $g \in N_G(\langle x \rangle)$.

□

Exercise 2.3.26. For a cyclic group C_n and each integer a, define $\sigma_a(x) = x^a$ for $x \in C_n$.

(a) Show that $\sigma_a \in \mathrm{Aut}(C_n)$ if and only if $\gcd(a, n) = 1$.

(b) Show that $\sigma_a = \sigma_b$ if and only if $a \equiv b \pmod{n}$.

(c) Show that every automorphism is σ_a for some a.

(d) Show $\sigma_a \circ \sigma_b = \sigma_{ab}$ and deduce $(\mathbb{Z}/n\mathbb{Z})^\times \cong \mathrm{Aut}(C_n)$, an abelian group of order $\varphi(n)$.

Solution. Write $C_n = \langle z \rangle$. Because C_n is abelian, the power map σ_a is a homomorphism.

(a) Its image is $\langle z^a \rangle$; it is surjective, hence an automorphism since C_n is finite, exactly when z^a has order n. This is equivalent to $\gcd(a, n) = 1$.

(b) Two homomorphisms from a cyclic group agree precisely when they agree on its generator. Thus $\sigma_a = \sigma_b$ if and only if $z^a = z^b$, or $n \mid (a - b)$.

(c) If $\tau \in \mathrm{Aut}(C_n)$, the generator z maps to another generator z^a. For all integers k, $\tau(z^k) = \tau(z)^k = z^{ak} = \sigma_a(z^k)$, so $\tau = \sigma_a$.

(d) For every $x \in C_n$,

$$(\sigma_a \circ \sigma_b)(x) = \sigma_a(x^b) = x^{ab} = \sigma_{ab}(x).$$

The assignment $[a]_n \mapsto \sigma_a$ from $(\mathbb{Z}/n\mathbb{Z})^\times$ is well defined by (b), maps to automorphisms by (a), is surjective by (c), injective by (b), and preserves multiplication by the displayed identity. Hence it is an isomorphism. Because $(\mathbb{Z}/n\mathbb{Z})^\times$ is abelian and has $\varphi(n)$ elements, so does $\mathrm{Aut}(C_n)$.

□